Cell

Establishing or Exaggerating Causality for the Gut Microbiome: Lessons from Human Microbiota-Associated Rodents

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https://doi.org/10.1016/j.cell.2019.12.025

Human diseases are increasingly linked with an altered or "dysbiotic" gut microbiota, but whether such changes are causal, consequential, or bystanders to disease is, for the most part, unresolved. Human microbiota-associated (HMA) rodents have become a cornerstone of microbiome science for addressing causal relationships between altered microbiomes and host pathology. In a systematic review, we found that 95% of published studies (36/38) on HMA rodents reported a transfer of pathological phenotypes to recipient animals, and many extrapolated the findings to make causal inferences to human diseases. We posit that this exceedingly high rate of inter-species transferable pathologies is implausible and overstates the role of the gut microbiome in human disease. We advocate for a more rigorous and critical approach for inferring causality to avoid false concepts and prevent unrealistic expectations that may undermine the credibility of microbiome science and delay its translation.

Introduction

Publications on the microbiome have increased rapidly over the last decade, with over 15,000 appearing in 2019 alone. Links with a wide range of debilitating chronic human diseases have considerable therapeutic implications and have attracted keen interest among scientists. Experimental animal models have provided strong empirical evidence for the role of the microbiome in chronic disease states (Atkinson and Chervonsky, 2012; Huttenhower et al., 2014; Maruvada et al., 2017). However, causal inferences are more elusive in humans (Lynch et al., 2019). Numerous human diseases have been associated with a fecal microbiota that is, in some aspects, different from that of healthy individuals, including changes in taxonomic composition, diversity indices, and/or functional readouts (Janssens et al., 2018). Such disease-associated microbiome alterations are often referred to as a "dysbiosis," a term that is widely used in the microbiome field but remains vaguely defined and is often misused (Hooks and O'Malley, 2017). Because a "normal" or "healthy" microbiome has not been defined yet (McBurney et al., 2019), we use the term dysbiosis to refer solely to an altered microbiome associated with a specific disease or condition as compared to a control. Whether such "dysbiotic" microbiomes are the cause or consequence of disease, or whether both are caused by a third factor, is, in most instances, unproven (Cani, 2019; Duvallet et al., 2017; Ni et al., 2017; Schmidt et al., 2018). Confirmation is essential if the microbiome is to become a valid target for interventions (Fischbach, 2018).

The challenge of establishing causality is not new to science or medicine. Transmuting epidemiological observations from correlation to causation has well-known limitations, and it is critical to be wary of correlations without mechanistic insights (Shanahan and Hill, 2019). There are two well-documented diseases in the microbiome field that link a microbial biomarker with causation in disease: *Helicobacter pylori*-associated peptic ulceration and gastric cancer (Parsonnet et al., 1991) and *Clostridium* (or *Clostridioides*) *difficile* infection-associated diarrhea (Gupta et al., 2016). However, causal inferences between complex microbiomes and other inflammatory, metabolic, neoplastic, and neuro-behavioral disorders have been neither compelling nor conclusive (Duvallet et al., 2017). Moreover, identification of causal components of complex microbiomes responsible for pathologies is complicated by the fact that microbiomes are highly individualized communities of bacteria, archaea, fungi, viruses, and protozoa, all of which along with their metabolites could play a role in disease, either alone or in concert. In addition, some microbiome components may protect or mitigate disease, and their influence may be context specific and may not occur directly but rather predispose the host to disease (Neville et al., 2018). Thus, causal relationships are seldom simple but rather complex and multi-dimensional and might be influenced by synergistic interactions within microbiomes.

Arguably, the most persuasive experimental evidence for the causal role of the gut microbiota in human disease can be achieved by determining the therapeutic value of fecal microbiota transplantation (FMT) (de Groot et al., 2017), which established the role of the gut microbiome in preventing *C. difficile*-associated disease (Gupta et al., 2016). Although an excellent experimental paradigm, human FMTs have limitations. First, they can only be used to study the protective attributes of the microbiome for ethical reasons. Second, FMTs may fail to correct altered microbiomes, particularly in diseases where the microbiome is not severely disrupted (Walter et al., 2018). Finally, FMTs are impractical for assessing the contribution of the gut microbiota to the risk of developing chronic diseases in humans because of the need for long prospective studies. Therefore, indirect experimental models are needed to establish causal relationships between altered microbiomes and pathologic conditions.

The most commonly used model to make such causal inferences is the transplantation of fecal microbial communities from individuals with and without a disease or pathology into germ-free rodents, followed by a comparative analysis of pathological phenotypes in the recipient animals. This model is known as human microbiota-associated (HMA) (Arrieta et al., 2016) or humanized gnotobiotic (Turnbaugh et al., 2009) rodents. In this review, we argue that the exceedingly high proportion of positive studies making causal claims from the use of HMA rodents is implausible and likely stems from a combination of insufficient rigor in experimental designs, inappropriate statistical analyses, and bias. Such exaggeration of the microbiome's role in disease jeopardizes the credibility of microbiome science and may delay its translation. We then try to turn criticisms into constructive suggestions and experimental guidelines for the evaluation of causal claims in animal models and address alternative approaches for examining causality. We advocate for a shift in mindset for microbiome science where reports are evaluated and published because of their scientific rigor (even if negative) rather than their sensational appeal.

HMA Murine Models

HMA murine models are used to answer a variety of scientific questions. First, they can determine whether a human microbiota can induce a pathology in a rodent disease model and establish the underlying mechanism and whether the phenotype is influenced by environmental factors, such as diet (Quach et al., 2018; Reigstad et al., 2015; Turnbaugh et al., 2009). Second, they can be used to establish causal relationships between altered microbiomes and diseases or other pathological or physiological conditions of the human host (Figure 1). The latter application is most commonly used and has been applied to a wide variety of pathophysiological states, such as obesity (Ridaura et al., 2013), asthma (Arrieta et al., 2015), allergies (Feehley et al., 2019), inflammatory bowel disease (IBD) (Britton et al., 2019), colorectal cancer (Wong et al., 2017), Parkinson's disease (Sampson et al., 2016), depression (Kelly et al., 2016), malnutrition (Smith et al., 2013), and autism (Sharon et al., 2019). Moreover, HMA murine models represent a platform to apply integrated "omics" approaches to identify the causal components of altered microbiomes that drive disease and the mechanisms by which pathologies emerge.

Although HMA rodents are considered the gold standard (Round and Palm, 2018; Staley et al., 2017) and cornerstone (https://www.nature.com/collections/bhciihjhei) for establishing the causal role of microbiome alterations linked to human disease, they have significant limitations for this application. A substantial proportion of the taxa that comprise the human gut microbiome fail to colonize in the recipient animals (Zhang et al., 2017). Taxa that do colonize assemble into communities that differ substantially from those found in the human donors (Staley et al., 2017). Most importantly, the ecological factors (such as diet, lifestyle, disease phenotype, and human genotype) that have driven the dysbiosis in humans in the first place are absent in the recipient rodents, making it unlikely that disease-associated alterations are replicated (Arrieta et al., 2016). Taxa that colonize at sufficient levels may not engage in the evolutionarily routed host-microbe interactions that they establish with their native host (Douglas, 2018). For example, HMA mice have deficits in immune, gut, and brain maturation (Chung et al., 2012) and differences in the metabolic interrelationship with the host (Marcobal et al., 2013). The relationship between a non-adapted gut microbiome and a surrogate host is likely to differ from that of a microbiome and host that have co-evolved over millennia.

Studies of bacterial pathogenicity demonstrate the difficulties of crossing host species. Consider *H. pylori* and *C. difficile*, two members of the human microbiome with established causality in human disease, neither of which exhibit sufficient colonization to cause reliable pathology in conventional murine models. Murine models perform well with naturally occurring pathogens that share virulence mechanisms with cognate human pathogens, but few murine models closely mimic human disease with a natural human pathogen. For example, the human pathogens enteropathogenic *Escherichia coli* (EPEC) and enterohemorrhagic *E. coli* (EHEC) mimic the mouse pathogen *Citrobacter rodentium*, in that they share a similar type 3 secretion system and associated type 3 effectors, yet only the latter causes disease in mice (Collins et al., 2014). Similarly, *Salmonella enterica* serovar Typhimurium (or *S.* Typhimurium), which causes limited intestinal diarrhea in humans, causes systemic invasive disease (liver and spleen involvement) in mice that mimics typhoid fever in humans (Coombes et al., 2005). Although murine models may be sufficient to study non-specific virulence factors (such as proteases encoded by *Pseudomonas aeruginosa*), they are seldom suited to decipher the exact disease mechanisms of specific human pathogens. Human pathogens, such as *Campylobacter jejuni*, *Salmonella enterica* serovar Typhi, *Vibrio cholerae*, *Shigella* species, and *Treponema pallidum*, neither colonize well nor replicate human-like disease in mice (Sawasvirojwong et al., 2013). In summary, although HMA murine models are conceptually appealing for studies of disease causation, the inability of human pathogens to replicate their pathologies in mice casts doubt on whether whole microbiomes can do so, especially because there are additional ecological limitations that would not arise with single pathogens.

A Systematic Review: HMA Murine Models Always Work!

We conducted a systematic literature review in June 2019 (with an updated literature search in October 2019) to assess the "success" rate for HMA murine models establishing or implying

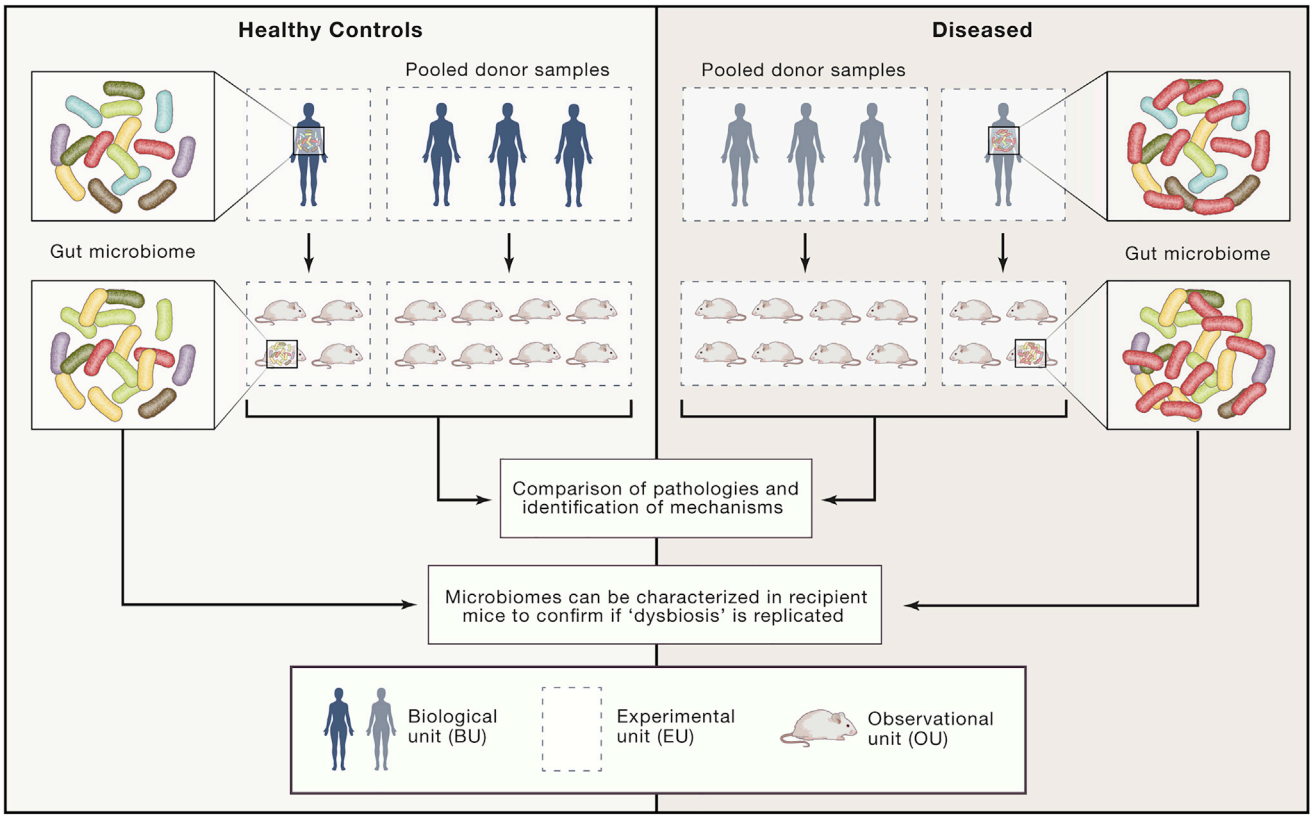


Figure 1. Experimental Design of HMA Murine Models

The overarching concept is to establish whether the altered or "dysbiotic" gut microbiota is associated with a human pathology by establishing it in germ-free or antibiotic-treated rodents and compare pathological phenotypes with animals that received microbiomes from healthy controls. The individual human represents the biological unit (BU), but donor fecal samples are often pooled, which reduces the number of experimental units (EUs). The EU establishes the "*N*" of the study and should serve as the unit of statistical inference (Lazic et al., 2018). The individual animals represent observational units (OUs). Characterization of the recipient rodents allows for the determination of whether the pathology can be transferred, as well as the underlying molecular mechanisms that drive the disease. In addition, comparisons of the gut microbiota in the recipient rodents with the microbiotas of the human donors can confirm that the dysbiotic patterns associated with the human disease have been replicated in the animals.

a causal relationship between altered human microbiomes and disease or physiological conditions (see Box 1 and Figure Box 1 for search strategy). Of 38 studies meeting the inclusion criteria, all but two (36/38; 95%) concluded that fecal transfer from diseased donors resulted in at least one phenotype of the human disease being greater in the recipient mice when compared with transfers from healthy donors (Figure 2; Table S1). The two exceptions, both of which studied the pathogenicity of the human microbiome in colorectal cancer, concluded that disease in the recipient mice was independent of the disease state of the donor but either dependent on the baseline microbiome of the mouse model (Baxter et al., 2014) or caused by microbes from human colon mucosal biofilms (Tomkovich et al., 2019). Thus, even in these negative studies, the disease in the recipient model was still causally linked with the human microbiome.

It is noteworthy that the studies identified in the systematic review covered a broad range of pathological processes: inflammatory; allergic; neoplastic; degenerative; metabolic; and developmental, many of which underpin chronic diseases with multifactorial causes with heritable and environmental components. Given this complexity and the fact that most reported "dysbiotic patterns" in human microbiomes are not disease specific and, therefore, are likely secondary to disease (Duvallet et al., 2017), causal claims of the microbiome seem implausible in all of these disease states. This prompts one to question whether the scientific criteria for human-like diseases in HMA murine models were sufficiently stringent.

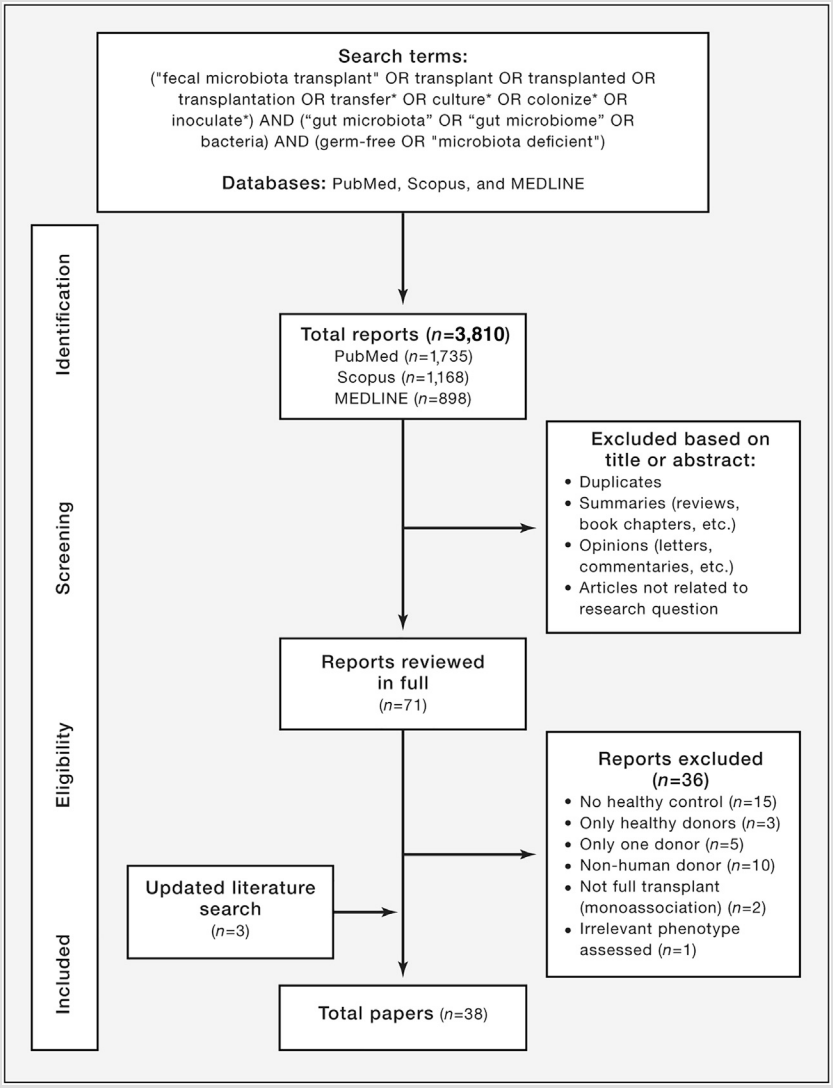
Lack of Mechanistic Insight

It is encouraging that 87% (33/38) of the studies in this systematic review addressed whether the gut or fecal microbiota in the rodents receiving microbiomes from healthy versus diseased individuals were actually different. However, only 63% (24/38) of the studies tested for a dysbiosis in the original human donor samples, and only 29% (11/38) confirmed that at least some aspect (reduced diversity, ecosystem processes/services, loss or bloom of specific taxa, or shifts in metabolic capacity) of the "human dysbiosis" was replicated in the recipient animals (Figure 2). Consequently, the majority of studies did not attempt to identify the "causal component" of the microbiome. Only a few studies (34%; 13/38) identified potential underlying

Box 1. Search Strategy for the Systematic Literature Review of HMA Murine Models

In order to determine the overall success rate in the published studies that utilized HMA murine models to establish whether gut microbiome alterations contribute to the etiology of human diseases, we conducted a systematic review of the literature in June 2019 (see Figure Box 1 for search strategy). We only included studies that compared pathophysiological phenotypes in animals (germ free or antibiotic treated) that received donor fecal samples from human individuals affected by a pathology versus controls. An updated literature search was conducted in October 2019 to include studies that had been published since the initial searches, and the same search strategy was used. In total, 38 studies were included in this review for analysis that assessed the role of the gut microbiome in the etiology of a number of disease states, pathologies, or physiological changes: allergies (Feehley et al., 2019; Petursdottir et al., 2017); *C. difficile*-associated diarrhea (Battaglioli et al., 2018); colorectal cancer (Baxter et al., 2014; Tomkovich et al., 2019; Wong et al., 2017); mental disorders (Fujii et al., 2019; Kelly et al., 2016; Zheng et al., 2016, 2019); HIV (Li et al., 2019); undernutrition (Kau et al., 2015; Smith et al., 2013; Wagner et al., 2016); IBD and irritable bowel syndrome (Britton et al., 2019; Chen et al., 2018; De Palma et al., 2017; Ge et al., 2017; Nagao-Kitamoto et al., 2016; Natividad et al., 2015; Torres et al., 2020; Touw et al., 2017); multiple sclerosis and Parkinson's disease (Berer et al., 2017; Cekanaviciute et al., 2017; Sampson et al., 2016); autism spectrum disorder (Sharon et al., 2019); arthritis and rheumatoid arthritis (Maeda et al., 2016; Stoll et al., 2019); frailty (Fielding et al., 2019); pregnancy-induced physiological changes (Koren et al., 2012); and obesity and its associated pathologies (i.e., metabolic syndrome, hypertension, and non-alcoholic fatty liver disease; Chiu et al., 2017; Goodrich et al., 2014; Kovatcheva-Datchary et al., 2015; Li et al., 2017; Liu et al., 2019; Ridaura et al., 2013; Soderborg et al., 2018; Zhang et al., 2017). Information on each study in this review is included in an Excel data extraction table in the supplementary information (see Table S1).

Flow Diagram of Article Search and Selection Process



The literature search was conducted in June 2019, and an updated literature search was conducted in October 2019.

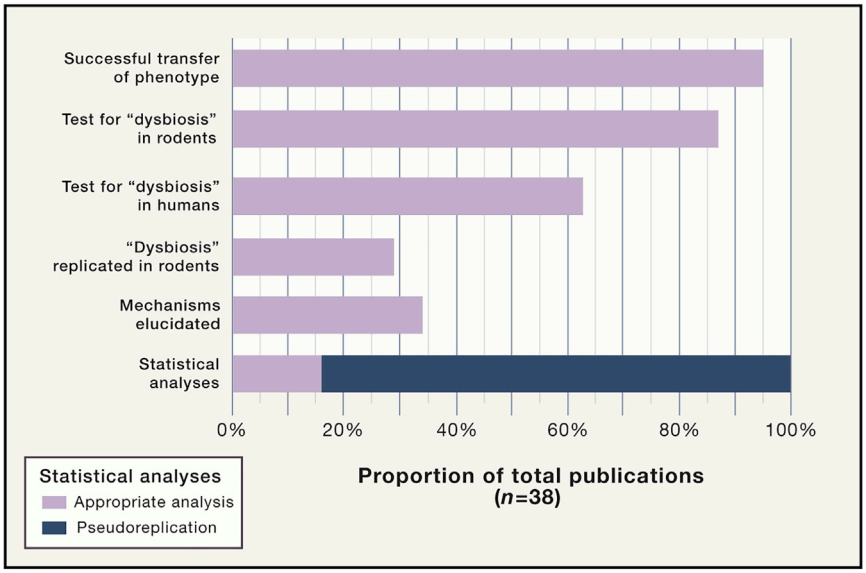


Figure 2. Results and Analyses Conducted in Studies Using HMA Murine Models

Studies were included in this analysis if both diseased and healthy control human donor samples were transplanted into germ-free rodents to assess the role of the altered gut microbiome in disease etiology. Data are reported as a percentage of all studies (n = 38). Studies were categorized as showing a successful transfer of the diseased phenotype if at least one aspect or symptom of the human disease or phathological phenotype was significantly greater in the "diseased" recipient mice when compared with the healthy controls. For statistical analyses, if the experimental N was not the same as the number of donor samples transferred from the human population (i.e., appropriate analysis), then these studies were classified as conducting pseudoreplication. See also Table S1.

mechanisms linking the dysbiotic microbiome with disease. One example reported that the genes *Tgfbr3* and *Ror2* were downregulated in the ileum of germ-free mice colonized with the microbiota of infants with cow's milk allergy, whereas *Fbp1* was upregulated in mice inoculated with the microbiota of healthy controls (Feehley et al., 2019). Monocolonization of germ-free mice with *Anaerostipes caccae* upregulated these genes, and mice were protected against anaphylaxis. Overall, the majority of reported experiments did not attempt to gain mechanistic insight as to how an aberrant gut microbiome might trigger disease development. We acknowledge that such experiments are potentially elaborate and might exceed available resources of smaller research groups. However, the capacity to establish mechanisms and causal components is a particular strength of murine models, and such information is pivotal for progression of the microbiome field (Fischbach, 2018; Hooks and O'Malley, 2017; Lynch et al., 2019).

Rigor of Experimental Design and Statistical Analysis

HMA rodents are a relatively new model, and experimental designs have not yet been standardized. Often, a small number (1–5) of human donors is used per human disease and control group (Kau et al., 2015; Kelly et al., 2016; Kovatcheva-Datchary et al., 2015; Ridaura et al., 2013; Smith et al., 2013; Soderborg et al., 2018), and these "donor microbiomes" are then replicated in a larger number of individual mice (Figure 1), which are then often used for statistical inference. Such an experimental design and analysis is questionable for several reasons. First, the small number of donors used in most studies does not capture the extensive inter-individual variability of the human gut microbiome, which is typically much larger than the effect sizes caused by disease states (Lloyd-Price et al., 2019; Namasivayam et al., 2019). A large proportion of this variability (>80%) cannot be attributed to known factors (Rothschild et al., 2018; Wang et al., 2016), and much of it might be caused by stochastic ecological processes (Martínez et al., 2018). Therefore, even if microbiome differences were causal in human disease, a small group of individuals would be unlikely to be sufficiently representative to reflect such changes.

Second, experimental effects in biology are established by independent replication of the individual units within the groups that are compared, which determines the sample size (N) and forms the basis of statistical inference (Lazic et al., 2018). It is the genuine replication of the "experimental unit" (EU) (the entity about which inferences are made) that increases the sample size (N) and, therefore, allows for the testing of experimental hypotheses. For HMA murine models, the EU is ideally identical to the biological unit of interest (BU), which is the human donor (Figure 1), because the purpose of the experiment is to determine the contribution of the gut microbiome to disease pathology in the human hosts. The individual rodent used for each experiment is merely an "observational unit" (OU), which is the entity on which measurements are taken. If only OUs and not EUs are replicated and used for statistical analysis as individual units, the result is "pseudoreplication," which artificially inflates the sample size, the chance for "false positive" findings, and the evidence for a scientific claim (Lazic et al., 2018). This problem is amplified if donor samples are pooled before the inoculation of groups of rodents, which is common practice for both the "diseased" and healthy control samples (Chen et al., 2018; Kelly et al., 2016; Koren et al., 2012; Petursdottir et al., 2017; Zheng et al., 2016), as it reduces the number of EUs to the number of inocula (Figure 1). We found that 84% (32/38) of the published studies on HMA murine models used the individual animals as the statistical inference, although animal numbers were much higher than the number of human donors and therefore inflated (Figure 2). Only 16% of studies (6/38) were clear in their use of the individual human donor samples as the "N" for statistical inferences or used the same number of rodents as donors.

In summary, the majority of the published studies on HMA rodents lack rigor in their experimental design and statistical analyses, as they use both an insufficient number of human donors and suffer from "pseudoreplication." It is noteworthy and encouraging that several recent studies using HMA rodents have applied appropriate statistics (Britton et al., 2019; Goodrich

et al., 2014; Li et al., 2019; Poole et al., 2019; Stoll et al., 2019). Another study has focused on the ecological aspects of the HMA animal model (Zhang et al., 2017). Although the authors showed differences in weight gain in mice receiving microbiomes from obese humans versus controls, metabolic phenotypes in the mice differed by isolators, with one control subgroup showing no differences in weight gain to the mice in the obesity group, although the other control group was clearly lower. Spread of microbes between cages within isolators contributed to the transmission of metabolic phenotypes. The findings highlight not only the importance of donor variability but also how the outcomes of HMA rodent experiments can be impacted by the environmental transmission of microbes, both of which have direct relevance for the design of such studies. Although there are undoubtable logistical and cost implications for the design of HMA animal studies, if the question is causality of an altered microbiome linked to a human phenotype, there is no substitute for an adequate replication and the prevention of microbial transfer between mice from different donors. If, however, the question is restricted to causality of the microbiome in general, or the sole study of mechanisms, *N* (BU) becomes less critical than the number of murine replicates (OU).

Where Are the Missing Negative Studies?

Although flaws in experimental design and statistical analysis are each likely to contribute to the overrepresentation of positive studies in our systematic review, there are less tangible but perhaps greater contributory factors. Bias arises at various levels: technical; experimental; analytical; interpretative; and at publication. Arguably, the biggest factor is that, in order to get a study published in a major journal, the results have to be positive, and negative findings are rarely published in these journals (Nissen et al., 2016) but end up in journals of lower prestige than those featuring the original claims (Matosin et al., 2014). Consequently, investigators (this includes the authors of this review) are incentivized to withhold data that do not support their hypotheses (the file drawer effect; Scargle, 2000) or to recast it in a way where negative findings are presented as positive results. This generates a bias toward positive studies. Negative data and failures to reproduce positive reports frequently go unreported and are not accessible for systematic review (Fanelli, 2012). Positive-result bias is not specific to microbiome science; it is a well-established, multifaceted phenomenon caused by all stakeholders (i.e., researchers, institutions, editors, journals, the media, and the public) in pursuit of the spectacular (Begley and Ioannidis, 2015). The problem might be especially hazardous for research on HMA animal models because of their centrality to the essential question of causality and because of their susceptibility to false positives (*vide supra*). When virtually all reports are positive, as in our systematic review, the greater the doubt will be and the more justified the scrutiny should be.

Why Is This Important?

Self-correction is probably the strongest facet of science, but it can only work if negative findings are published (Nature, 2017; Nissen et al., 2016). Clinical translation of microbiome science requires confirmation of causal host-microbe relationships along with identification of contributory mechanisms (Fischbach, 2018). Specifically, the correction of altered microbiomes, be it by FMT, live biotherapeutics, or microbial products, will be beneficial only if the alterations are causal or contributory to the disease rather than a bystander response. Indeed, some alterations in the microbiome might be a compensatory beneficial response to disease, and their "correction" might be detrimental. There are, therefore, public health implications from misstating the evidence for causality of the microbiome in human disease. Concerns surrounding inappropriate use of FMT for conditions other than recurrent *C. difficile* infection are testimony to the importance of the accuracy of messages from microbiome scientists.

In addition to rigorous analysis, the language used to report results is an important modifier of the message. Human phenotypes transferred to recipient animals in HMA rodent experiments are proxies for human diseases that do not occur naturally in rodents. In this respect, what precisely does autism-like behavior, depression, or anxiety mean in an animal (Belzung, 2014; Freudenberg et al., 2018; Kazdoba et al., 2016; Steimer, 2011)? Criteria for "human-like" disease are vague, often undefined, or simply described *post hoc*, and therefore, it is critical that interpretations should be limited to pathophysiological and behavioral mechanisms and not the clinical syndromes in humans. Elsewhere, we have argued that hype and hyperbole should not detract from the success stories of microbiome science and the pace with which they are being translated into clinical diagnostics, prognostics, and therapeutics (Shanahan, 2015). However, causal claims in the microbiome field are often overstated and disproportional to the experimental evidence from which they are derived (Hanage, 2014; Lynch et al., 2019), and it has become difficult to identify a disease state in which the microbiome has not been implicated in the scientific literature (Duvallet et al., 2017), a notion amplified in the popular press.

How to Improve Experimental Rigor when Testing for Causality

HMA rodents are powerful models for studying both causality and mechanisms of host-microbe interactions, and we do not discourage their use. They are particularly suited to study metabolic host-microbiome interactions (e.g., as it relates to differences in diet), even if colonized by human microbes that are not perfectly adapted to rodents. They are less reliable for interrogating immune-mediated disorders, where transferred phenotypes are often poor replicas of the human condition (Chung et al., 2012). To test for the causal role of microbiome alterations associated with human disease, HMA animal experimental designs have to be robust. We have extended our previous suggestions on how HMA murine models should be designed, analyzed, and interpreted (Arrieta et al., 2016) and summarize our recommendations in Box 2. Methodologies and standards will likely continue to evolve, and we encourage others to propose alternative or additional guidelines and strategies to improve these outlined recommendations that are meant to act as a starting point for a discussion on how causality of microbiomes should be assessed. The field would benefit from guidelines of an expert consortium that define minimal information required from experiments using HMA animals, as established by other disciplines in other contexts (Bowers et al., 2017;

Box 2. Recommendations for HMA Murine Models to Establish Causal Relationships between Altered (Dysbiotic) Gut Microbiomes and Human Disease, Pathophysiological State, or Metabolic Condition

1 To establish causal claims in HMA rodents, compare donor samples from individuals with a disease (or pathophysiological state) with those that are healthy (or those that do not show the respective phenotype) and try to determine the microbiome alterations (dysbiotic patterns) associated with the pathology.
2 Use an appropriate number of donors to account for real biological variation. Sample size will depend on the magnitude of the effects and error rates, which are unknown in most cases and will be dependent on the specific animal model and the human pathology studied. Power analyses are, therefore, difficult to perform but will become possible once more data become available and should be included if possible.
3 Avoid pseudoreplication. Use donor numbers as "*N*" for statistical inferences instead of the number of recipient animals for statistical analyses. To ensure proper replication, prevent spread of microbes between mice from different donors. This can be achieved by the use of ventilated racks and appropriate barrier husbandry practices for cage changing, sanitation, and animal handling (Franklin and Ericsson, 2017).
4 Do not pool donor samples, as this eliminates the variation between individuals and reduces the number of "experimental units" to the number of inocula used. If donor samples are pooled, consider that this reduces the number of experimental units and, with this, the *N* for statistical inferences.
5 Confirm whether microbiome engraftment was successful and whether "dysbiotic" patterns were transferred from the human cohort to the recipient animals. This analysis should focus on composition, diversity, functional potential (pathways), and functional output (metabolites and bacterial compounds) through a multi-omics analysis. In addition, establish the correct time frame in which experiments are conducted, as it can take considerable time to establish and/or influence phenotypes in the recipient animals. If possible, use cryopreservation for optimal storage of fecal donor samples (Vandeputte et al., 2017).
6 Provide an honest discussion of the limitations of animal models and the findings of the study in a publication and avoid overstatements on causal claims, especially for pathological phenotypes that are just proxies for the human disease states.
7 Longer term goals: attempt to gain mechanistic insight. Identify candidate "causal components" of the gut microbiota (specific taxa, causal core, ecosystem process/service, overall diversity, metabolic pathways and functions, metabolites, etc.) by correlating findings from multi-omics analysis with host phenotypes. Follow up with functional experiments (e.g., in gnotobiotic rodents) to confirm causal components and decipher mechanisms by which pathologies emerge. Such mechanistic studies are often dauntingly complex and beyond the scope of a single publication but should be pursued, perhaps through funded initiatives, due to their pivotal importance for the microbiome field.

Brazma et al., 2001). Grants should be written to allow for experiments that are adequately powered and designed, and funding bodies should be prepared to accept greater costs to allow such experiments. Criteria for accepting phenomenology in recipient rodents as evidence for a role in human disease need greater scrutiny and rigorous precision and should be preferably related to likely molecular mechanisms (Shanahan and Hill, 2019). Given the importance of causal inferences and mechanisms (Fischbach, 2018), an improvement of the standards by which HMA animal studies are performed might contribute to solving problems that have been identified for the microbiome field, such as positive-research bias, hype, lack of scrutiny, language, definitions, etc. (Bourrat, 2018; Brüssow, 2019; Lynch et al., 2019; Shanahan and Hill, 2019).

Additional strategies for improving HMA animal models deserve consideration (Figure 3). In view of the limited capacity of rodents to mimic humans, rodents that are immunologically or metabolically "humanized" (e.g., by engraftment with human immune or metabolic cells) will increase the likelihood of detecting host-microbe interactions specific to humans (Figure 3A; Garcia and Freitas, 2012; Henderson et al., 2019; Walsh et al., 2017). Alternatively, experimental hosts that are better mimics of human physiology, such as pigs and primates (Dhakal et al., 2019; Heinritz et al., 2013; Nishida and Ochman, 2019), merit greater usage for conditions for which these animals are improved models (Figure 3B) if their greater cost can be constrained. To establish a framework to test for causation in human microbiome research, Koch's postulates have been revised and experimental procedures have been proposed to validate hypotheses inferred from genomic analyses of microbiomes (Byrd and Segre, 2016; Neville et al., 2018). These frameworks capitalize on recent breakthroughs in the identification of candidate causal components by interrogating multi-omics datasets, their isolation into pure cultures, and their study in animal models, specifically gnotobiotic mice in which microbiome membership can be strictly controlled (Round and Palm, 2018). Working with defined cultures would negate some of the strict requirements of replication that apply to HMA rodents as long as pathogenic or beneficial traits are not subject to strain-to-strain variation, and strains are used that are representative of the causal component to be tested (which can be a microbial species, subspecies, lineage, or type). Although powerful, these models suffer from some of the same limitations of HMA rodents, but these may be offset if used in more suitable recipient hosts (Figures 3A and 3B).

Given the limitations of extrapolating findings from animals to humans, approaches to establishing causal links directly in humans have emerged. Novel statistical approaches, such as Mendelian randomization (Sanna et al., 2019) and mediation analysis (Leong et al., 2018), have been used to make causal inferences between gut microbiome metabolites and host markers in human studies, and mathematical and machine learning approaches, such as causal molecular networks (Hill et al., 2016), structural equation modeling, or Bayesian network inference, could be adopted to microbiome data (Schmidt et al., 2018; Sohn and Li, 2019; Figure 3C). Strategies like mediation analysis benefit from longitudinal data in which the temporal order of variables can help disentangle cause-effect relationships, in that effects cannot temporally precede causes (Gelfand et al., 2009). However, whether the statistical assumptions necessary for these approaches to make reliable causal inferences can be made for the microbiome must first be established and validated.

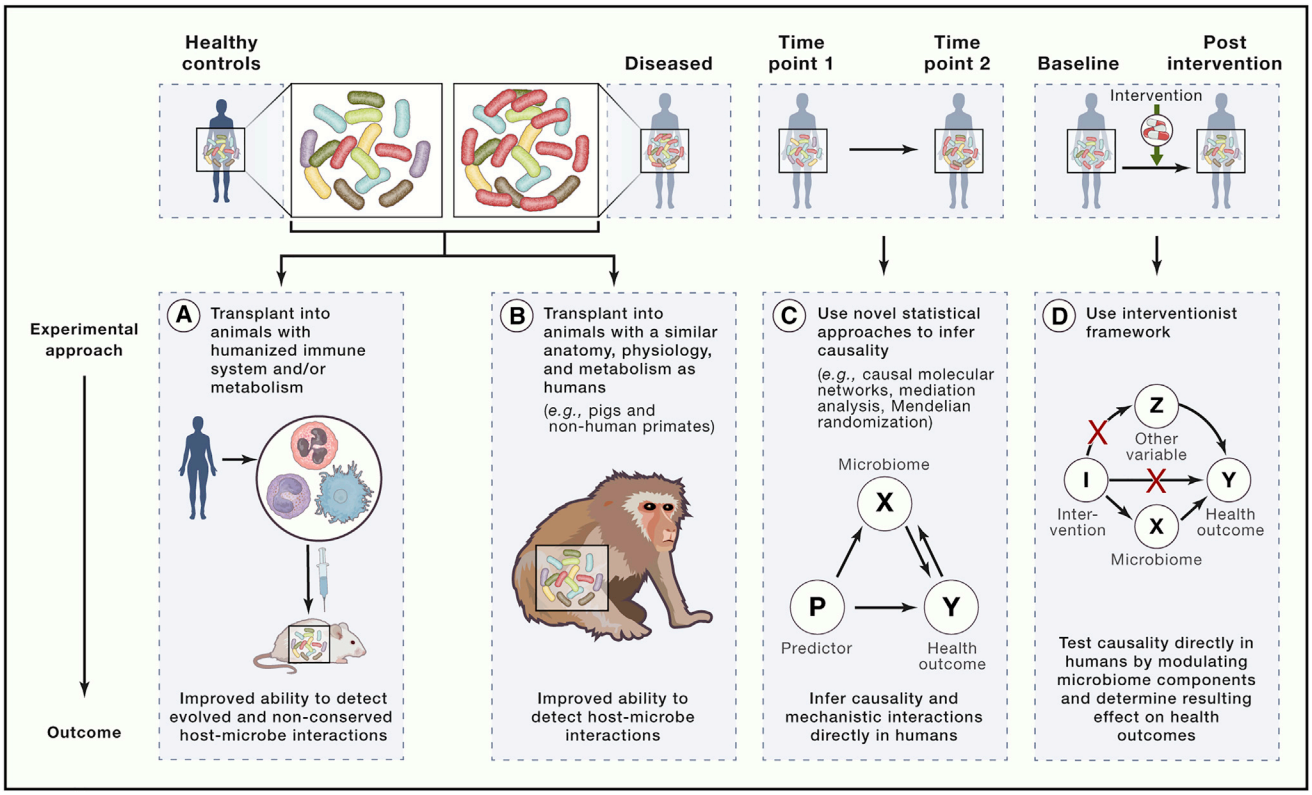


Figure 3. Alternative Experimental Approaches to Determine the Causal Role of the Gut Microbiome in Disease Etiology

(A) Rodents that are immunologically or metabolically "humanized" (e.g., by engraftment with human immune or metabolic cells) to increase the likelihood of detecting host-microbe interactions specific to humans.

(B) Use of experimental hosts that are anatomically, physiologically, and metabolically closer to humans.

(C) Novel statistical approaches, such as Mendelian randomization, mediation analysis, and mathematical and machine learning approaches (such as causal molecular networks, structural equation modeling, or Bayesian network inference), could be adopted to microbiome data.

(D) An "interventionist framework" could demonstrate cause-effect relationships by showing that interventions targeting the putative cause (the gut microbiome) produce changes in the effect (host phenotype).

Recently, Lynch and colleagues proposed the application of the "interventionist framework" by Woodward (2010) to the microbiome (Figure 3D), based on the idea that cause-effect relationships can be inferred when an intervention directed at a putative cause elicits a favorable effect (Lynch et al., 2019). Although this approach also has limitations (Woodward, 2016), it was successfully used to demonstrate *H. pylori* as a cause of peptic ulceration and the microbiome as a protective factor against *C. difficile*-associated disease. Future interventions could be more complex and directed at microbiome patterns or ecosystem processes linked with disease in cohort studies or population-level analyses (Arrieta et al., 2015; Falony et al., 2016; Liu et al., 2017; Yassour et al., 2016) but could also employ specific microbes and/or their molecules and metabolites (Olle, 2013). A proof of concept has been reported by targeting *Akkermansia muciniphila*, a bacterial species shown to be associated with improvements in metabolic health and weight management (Liu et al., 2017; Yassour et al., 2016), in a human intervention trial (Depommier et al., 2019). If these effects are independently replicated, causal claims for *Akkermansia* in humans become realistic. Enhanced precision in manipulation of the microbiome, by both additive (FMTs, microbes, and their molecules and metabolites) and subtractive approaches (bacteriophages and CRISPR-Cas antimicrobials) will broaden the potential for interventionist approaches, but this will require deeper understanding of the ecological constraints under which such manipulations can be achieved (Koskella et al., 2017; Martínez et al., 2018; Walter et al., 2018). The interventionist framework not only provides a platform to determine causality but also the underlying mechanisms if informed by clinical metadata and follow-up experimentation in animal models (e.g., improved models, such as those in Figures 3A and 3B).

Changing Mindsets and Policies

Microbiome science will obviously benefit from greater rigor and critique, but as with all sciences, there are changes needed elsewhere to fulfill its promise. Long overdue are changes to the incentives and criteria by which scientific endeavor is valued and judged. Detailed lists of corrective measures relevant to all areas of science have been proposed (Begley and Ioannidis, 2015), but overstatement due to publication bias seems to be particularly problematic in relation to the causal role of the microbiome in human disease. Some scientific journals have already published occasional "negative issues," but more preferable would be a

shift in attitude toward judging negative findings based on their scientific merit rather than their absence of sensationalism. In addition, negative findings should not be relegated to a single isolated issue of a journal. Over half a century ago, Karl Popper recommended that rigorous research should attempt to reject its hypotheses; "every genuine test of a theory is an attempt to falsify it, or to refute it" (Popper, 1963). Sadly, most researchers and reviewers in the microbiome field, and almost all editors of high-impact journals, seem to regard acceptance of the null hypothesis as a failure. As with all scientific endeavor, our field would be served better if editorial policies of higher tier journals and guidelines of funding agencies discouraged any discrimination against negative data, replications, validations of experimental designs, and statistical analyses. Better still, they should solicit such scrutiny (Begley and Ioannidis, 2015). Because scientific careers and research funding are gauged to a large degree by the success of investigators in these journals, changes in their policies could be transformative.

Conclusions

Pendulums swing both ways. Once was a time when the intelligentsia was slow to accept *H. pylori* as a cause of seemingly heterogeneous and complex diseases in humans. Currently, there seems to be wishful thinking that the microbiome contributes to most diseases and pathological processes, no matter how complex their etiology is. It is now time to revert toward a more critical and rigorous scientific approach for microbiome research, especially when using HMA rodents as its cornerstone model for establishing causality, which will ensure that this exciting field will capitalize on its promise to improve human health.

SUPPLEMENTAL INFORMATION

Supplemental Information can be found online at https://doi.org/10.1016/j.cell.2019.12.025.

ACKNOWLEDGMENTS

J.W. acknowledges support through the Campus Alberta Innovation Program, Canadian Institutes of Health Research, and the Weston Family Microbiome Initiative. B.B.F. acknowledges support through Canadian Institutes of Health Research, Gates Foundation, Genome British Columbia, and Canadian Institute for Advanced Research. F.S. is funded in part by Science Foundation Ireland Centre grant (APC/SFI/12/RC/2273).

AUTHOR CONTRIBUTIONS

J.W. and A.M.A. conceived the idea for this manuscript. A.M.A. conducted the systematic review and data extraction. All authors contributed to writing and finalizing this manuscript.

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